Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 5 [2011 P1]

Test Date: 20th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2011-P1-Model

Time: 3:00 **IMPORTANT INSTRUCTIONS** **Max Marks: 240**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 8 – 11)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 12 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 17 – 23)	Questions with Integer Answer Type	4	0	7	28
Total				**23**	**80**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 24 – 30)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 31 – 34)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 35 – 39)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 40 – 46)	Questions with Integer Answer Type	4	0	7	28
Total				**23**	**80**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 47 – 53)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 54 – 57)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 58 – 62)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 63 – 69)	Questions with Integer Answer Type	4	0	7	28
Total				**23**	**80**

CHEMISTRY: **Max.Marks: 80**

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

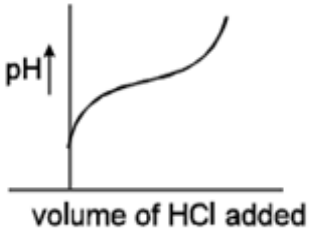
1. value of Λ_m^∞ for $SrCl_2$ in water at 25ºC from the following data:

Conc. (mol/lit)	0.25	1
$\Lambda_m\left(\Omega^{-1}cm^2mol^{-1}\right)$	260	250

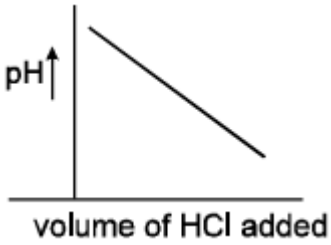
A) 270 **B)** 260 **C)** 250 **D)** 255

2. When 100 ml of 0.1 M NaCN solution is titrated with 0.1 M HCl solution the variation of pH of solution with volume of HCl added will be:

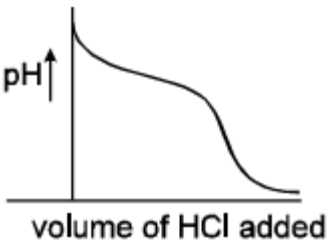
A)



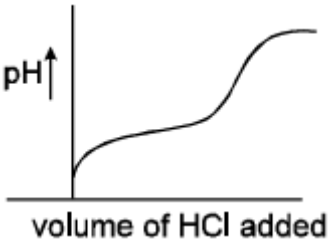
B)



C)



D)



3. During winters, moisture condenses in the form of dew and can be seen on plant leaves and grass. The entropy of the system in such cases decreases as liquids possess lesser disorder as compared to gases. With reference to the second law, which statement is correct, for the above process

A) The randomness of the universe decreases
B) The randomness of the surroundings decreases
C) Increase is randomness of surroundings equals the decrease in randomness of system
D) The increase in randomness of the surroundings is greater as compared to the ecrease in randomness of the system.

4. Indicate T for true and F for false for the following statements S_1, S_2, S_3 *and* S_4 respectively

S_1: Argon is used in arc welding of metals or alloys to provide an inert atmosphere.

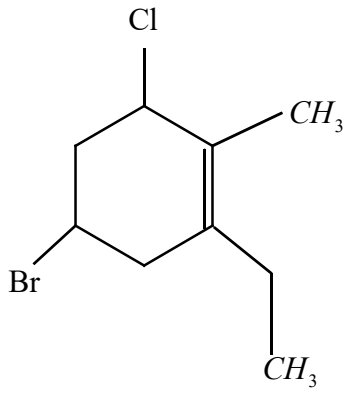
S_2: XeF_2, XeF_4 *and* XeF_6 are colourless crystalline solids and sublime readily at around 298K.

S_3: XeF_2, XeF_4*and* XeF_6 undergo hydrolysis with water.

S_4: Xenon fluorides react with fluoride ion acceptor to form cationic xenon species and with those of fluoride ion donors to form xenon fluoro anions.

A) FFFF **B)** TFTF **C)** TTTT **D)** FTFT

5. IUPAC name of:



A) 4-Bromo-6-chloro-2-ethyl-1-methylcyclohex-1-ene

B) 5-Bromo-1-chloro-3-ethyl-2-methylcyclohex-2-ene

C) 5-Bromo-3-chloro-1-ethyl-2-methylcyclohex-1-ene

D) 1-Bromo-5-chloro-3-ethyl-4-methylcyclohex-3-ene

6. What will be the total number of possible structural isomers with formula $C_7H_{13}Br$ fulfilling the following conditions

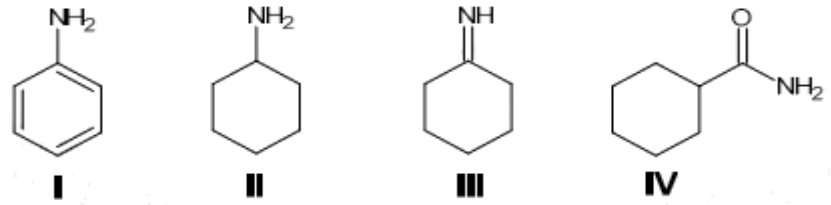
(I) it should contain 6 membered ring

(II) none of the isomer should contain both bromine atom either on same or adjacent carbon atom

(III) none of the isomer should contain bromine atom on tertiary carbon atom

A) 5 **B)** 8 **C)** 3 **D)** 7

7. Choose the correct order of basic character of the following compounds



A) I>II>IV>III **B)** II>I>III>IV **C)** II>III>I>IV **D)** II>I>IV>III

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

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8. The standard reduction potentials of some half-cell reactions are given below

$PbO_2 + 4H^+ + 2e^- \rightleftharpoons Pb^{2+} + 2H_2O \qquad E^o = 1.455V$

$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+}4H_2O \qquad E^o = 1.51V$

$Ce^{4+} + e^- \rightleftharpoons Ce^{3+} \qquad E^0 = 1.61V$

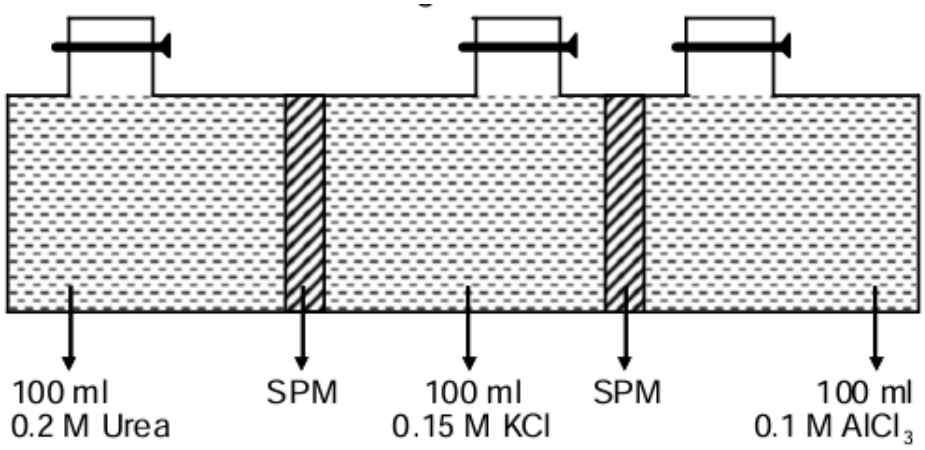
Pick out the correct statement:

A) Ce^{+4} will oxidize Pb^{2+} to PbO_2 **B)** MnO_4^- will oxidize Pb^{2+} to PbO_2

C) Ce^{+4} will oxidize Mn^{+2} to MnO_4^- **D)** PbO_2 will oxidize Mn^{+2} to MnO_4^-

9. Consider the following system.

Three different aqueous solution each having volume 100 ml are taken and kept in contact as shown.



After sufficiently longer time (Consider temp constant & 100% dissociation of strong electrolyte) the true statements regarding the above system

A) Volume of urea solution will be $\frac{100}{3} ml$

B) Volume of $AlCl_3$ solution will be $\frac{400}{3} ml$

C) There will be no change in volume of KCl solution.

D) Volume of both KCl and $AlCl_3$ solutions will increase.

10. Select the correct statements.

A) $\ln(Ph_3Si)_2O$ the Si — O — Si group is nearly linear.

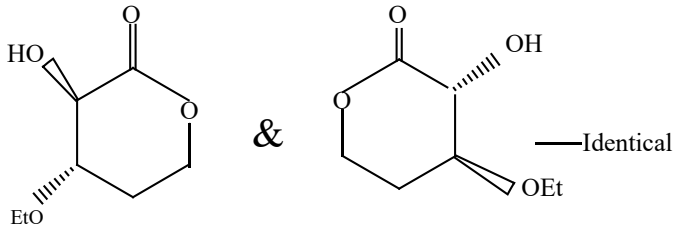
B) Silanols such as $(CH_3)_3$ Si-OH are stronger protonic acids than $(CH_3)_3$ C-OH

C) Trisilyl phosphine $(H_3Si)_3P$ is planar (not considering hydrogens on silicon)

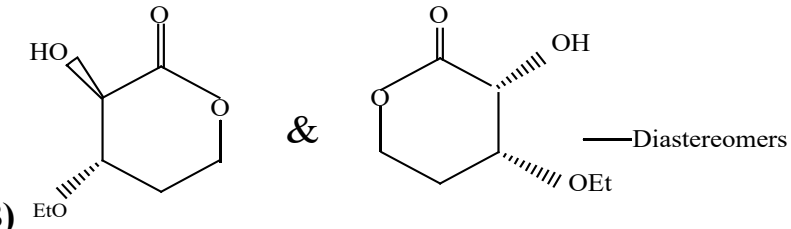
D) ClO_4^- does not polymerise due to strong π-bonds.

11. How many of the following is/are correctly matched?

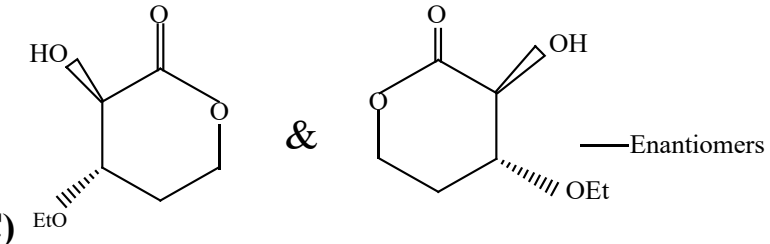
A)



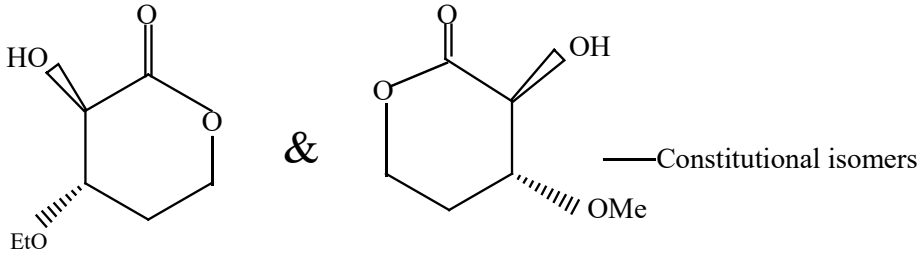
B)



C)



D)



SECTION – III

(PARAGRAPH QUESTIONS TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions and based on the other paragraph 2 multiple choice questions have to be answered. Each of these questions has four choice (A) , (B) , (C) and (D) out of which ON:Y ONE is correct

Passage – I:

The system shown in the figure is in equilibrium, where A and B are isomeric liquids and form an ideal solution at TK. Standard vapour pressures of A and B are $P_A^o \text{ and } P_B^o$ respectively at T=50K. The vapours of A and B are collected into two separate containers each of volume V, container-I is maintained at 2TK and container-II is maintained $\frac{3T}{2}$ K temperatures.

At the temperature greater than TK, both A and B exist only in gaseous form.

Total vapour pressure of the system at TK is given as $P = P_A^{\,o} X_A + P_B^{\,o} X_B$

where $X_A \text{ and } X_B$ are the mole fractions of A and B in the liquid mixture.

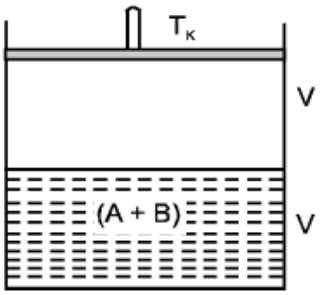
The collected vapours in container-I behaves ideally at 2TK where the isomerization reaction occurs in which A is converted into B by first order kinetics as $A \xrightarrow{K'} B$ (K' is rate constant)

In container-II at the temperature $\frac{3T}{2}$ K, A and B are ideal in nature and non-mixing. A small pin hole is made in the container-II. Where the initial rate of effusion (r) of both gases in vaccum can be determined by the expression $r = \frac{kp}{\sqrt{m}}$

Where p = pressure difference between system and surroundings

k = positive constant

m = Molecular weight of the gas.



12. If the partial vapour pressure of A is twice that of partial vapour pressure of B and total vapour pressure is equal to 2 at T = 50 K and V = 8.21 lit, then number of mole of A and B in vapour phase is

A) $\frac{8}{3},\frac{4}{3}$ **B)** 3,1 **C)** 2,2 **D)** 1,3

13. Vapours collected are kept in the container-I of volume 8.21 lit, maintained at 100 K and after 5 min the number of mole of B were found to be $\frac{8}{3}$. Then the pressure developed in the container after two half-lives is

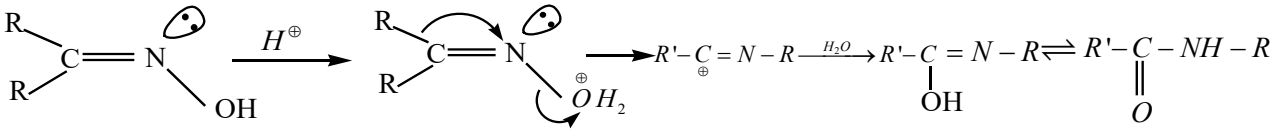
A) 2atm **B)** 4atm **C)** 1atm **D)** 0.5 atm

14. If vapour is collected in container-II of volume 8.21 lit maintained at 75 K then ratio of initial rate of effusion of Gas-A to Gas-B is

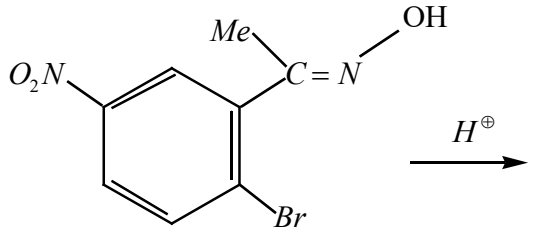
A) 2 : 1 **B)** 1 : 2 **C)** 4 : 1 **D)** 1 : 4

Passage – II:

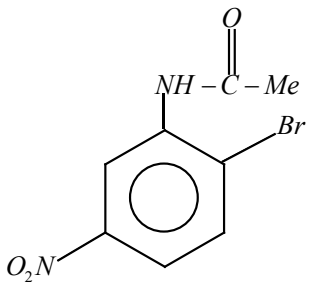
Aldehydes and ketones reacts with NH_2OH to form Aldoximes and Ketoximes respectively. Configuration of these can be determined by Beckmann rearrangement as that group migrates which anti w.r.t-OH



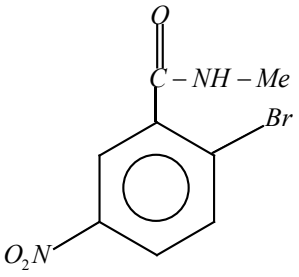
15.

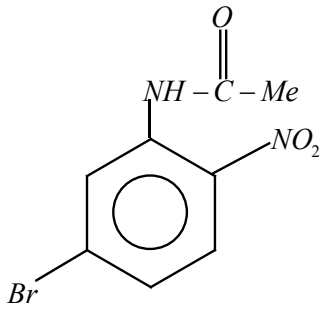


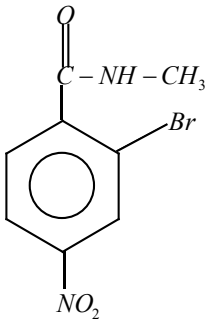
A)

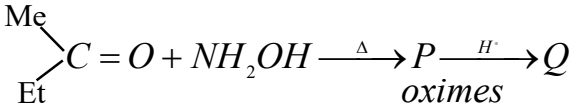


B)



C) 

D) 

16. 

Which of the following statement is/are incorrect?

A) oximes of P are geometrical isomers

B) Q are metamers

C) methyl is having less migrating aptitude than ethyl

D) Major product in Beckmann rearrangement is decided by shifting of group which is having more migrating aptitude

SECTION –IV

(INTEGER ANSWER TYPE)

This section contains **7 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

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17. In the acid base titration $H_3PO_4(0.1M)$ verses $NaOH(0.1M)$, the e.m.f of the solution is measured by coupling the electrodes with suitable reference electrode. When alkali is added pH of solution is in accordance with equation

$E_{cell} = E^o_{Cell} + 0.059\,pH$

The cell e.m.f. at the second equivalence point of the titration is____ volts

Given E^o_{cell} at this stage is 1.3805 V

(For $H_3PO_4\ ka_1 = 10^{-3}; Ka_2 = 10^{-8}; ka_3 = 10^{-13}$)

18. Depression in freezing point of aqueous FeC_2O_4 solution is_____ ^0C. (Round off to nearest integer)

When 10 ml of same FeC_2O_4 solution requires 25 ml of 0.4 M $KMnO_4$ in acidic medium to reach the equivalence point. Take molarity as molality, asume 100% ionization of FeC_2O_4 and neglect the hydrolysis of FeC_2O_4 $\left(k_f \text{ of } H_2O = 1.86\ ^0C\,kg\,mol^{-1}\right)$

19. Solid ammonium carbamate (NH_2COONH_4) alone was taken in excess in closed container of volume 5 Litre. Then the following equilibrium is setup

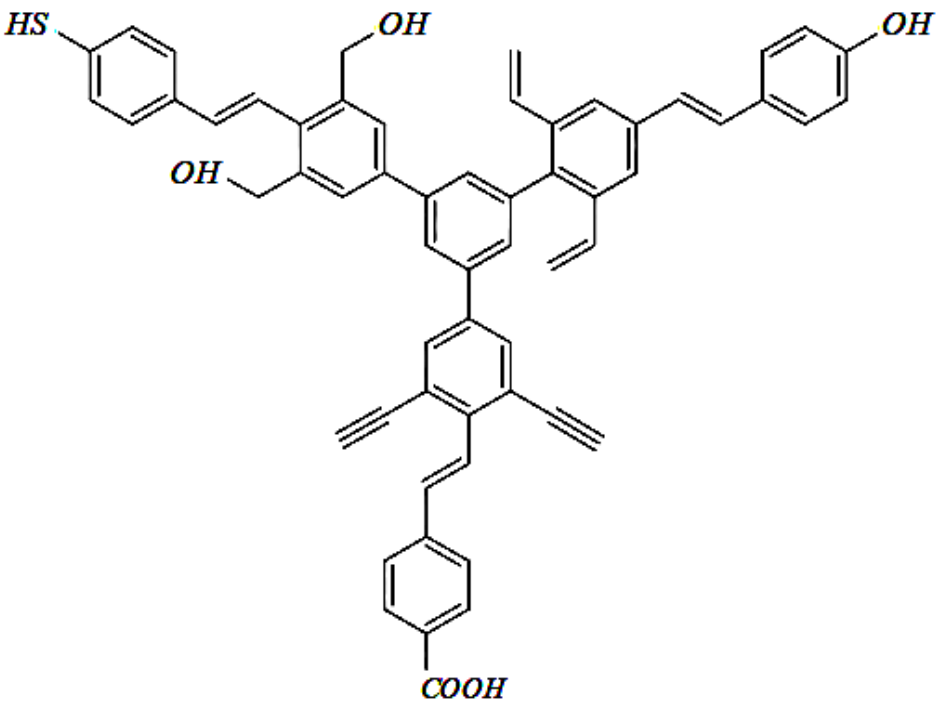
$NH_2COONH_4(S) \rightleftharpoons 2NH_3(g) + CO_2(g)$

equilibrium partial pressure of ammonia is 4 atm and equilibrium constant K_p is $x\,atm^3$ in 5 litre container when the above equilibrium mixture is transferred to another vessel of volume 10 litre then the total equilibrium pressure is y atm. Assuming temperature is constant during the whole experiment. The value of $\left(\frac{x+y}{10}\right)$ is_____(Round off to the nearest integer)

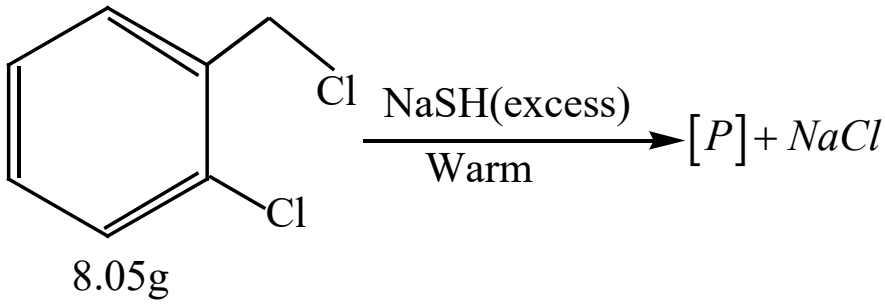
20. Among the following complexes, how many have 'spin only' magnetic moment of 2.83 B.M.?

$[Ni(CO)_4], [Ni(CN)_4]^{2-}, [NiCl_2(PPh_3)_2], [NiCl_4]^{2-}, [NiF_6]^{2-}, [NiF_6]^{4-}, [Ni(NH_3)_6]^{2+}, [Ni(en)_3]^{2+}, [Ni(H_2O)_6]^{2+}$

21. The number of moles of CH_4 produced when 1 mole of the following compound is treated with excess methyl magnesium bromide (CH_3MgBr) is_____



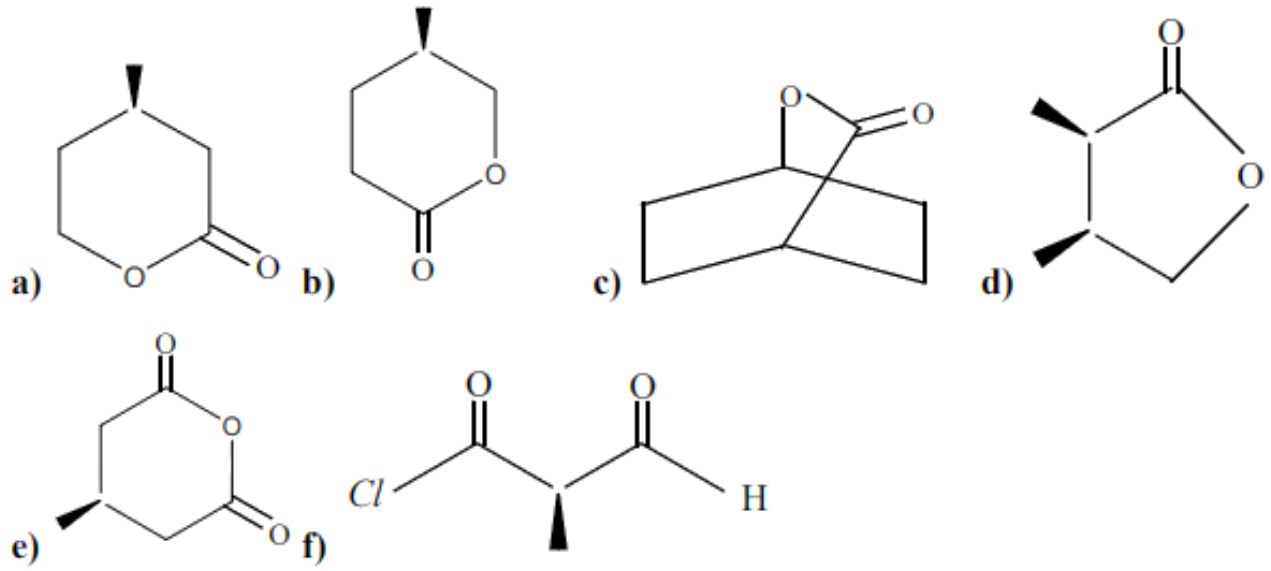
22. Consider the following reaction:



The mass (in g) of the organic product [P]is

[Atomic masses: H = 1, C =12, Na = 23, S = 32, Cl = 35.5] give your answer in the nearest integer value.

23. Find the number of compounds which on reduction with $LiAlH_4$ give optically inactive?



PHYSICS: **Max.Marks: 80**

SECTION – I

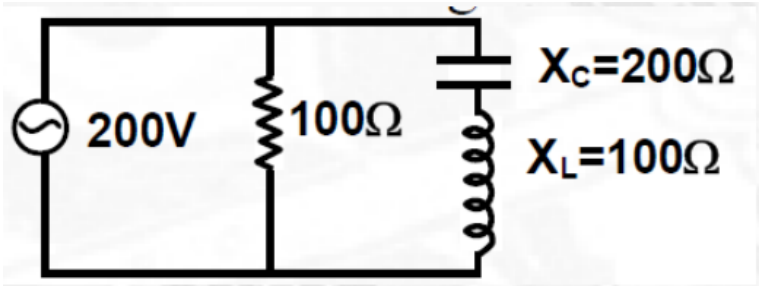
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

24. A point moves rectilinearly with deceleration which depends on the velocity v of the particle as $a = k\sqrt{v}$ where $k = +ve$ constant. At the initial moment the velocity of the point is equal to v_0. What distance will it cover before its stops.

A) $s = \frac{2v_0^{3/2}}{3k}$ **B)** $s = \frac{2v_0^{1/2}}{3k}$ **C)** $s = \frac{2v_0}{7k}$ **D)** $s = \frac{2v_0^{3/2}}{k}$

25. In the circuit shown, effective current through source is



A) 2A **B)** $2\sqrt{2}A$ **C)** 0.5A **D)** $\sqrt{0.4}A$

26. Which one of the following statements is WRONG in the context of X - rays generated from a X-ray tube?

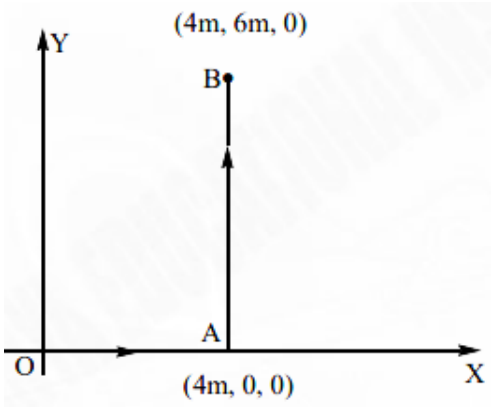
A) Wavelength of characteristic X-rays decreases when the atomic number of the target increases

B) Cut-off wavelength of the continuous X-rays depends on the atomic number of the target

C) Intensity of the characteristic X-rays depends on the electrical power given to the X-ray tube

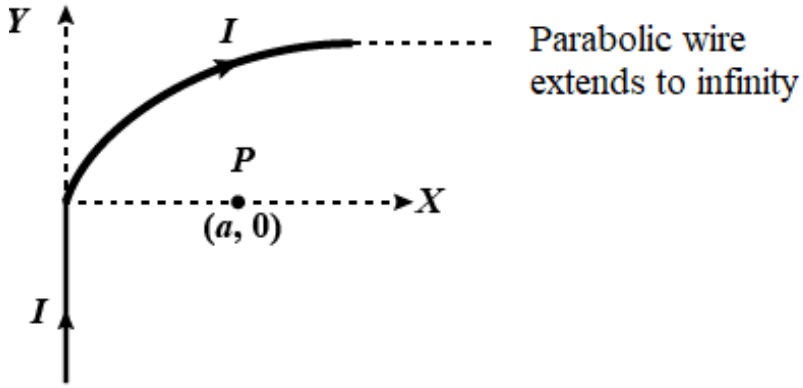
D) Cut-off wavelength of the continuous X-rays depends on the energy of the electrons in the X-ray tube

27. Find the work done (in Joule) by a force, $\vec{F} = \left[\left(y^2 - x^3 + 5z^2\right)\hat{i} + \left(3xy - 2z\right)\hat{j} + \left(3z\right)\hat{k}\right]$ newton in taking a particle from origin O to the point 'B' (4m, 6m, 0) along the path OAB as shown in the figure



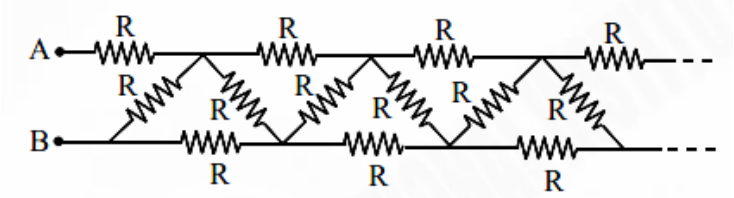
A) 64 **B)** 256 **C)** 216 **D)** 152

28. An infinitely long wire bent into shape as shown in figure carries a steady current I. If the curved wire is along the curve $y^2 = 4ax$, then magnetic field at point P(a ,0), is



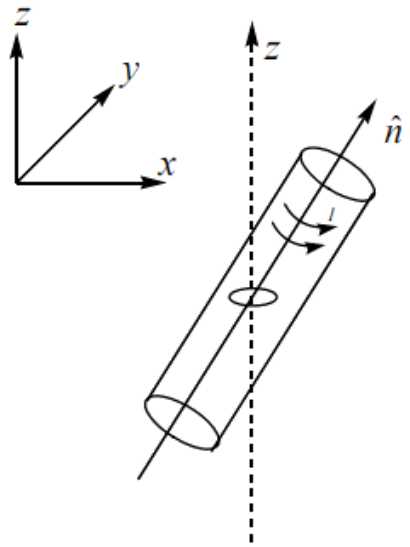
A) $\frac{\mu_o i}{4\pi a}$ **B)** $\frac{\mu_o i}{8\pi a}$ **C)** $\frac{\mu_o i}{4a}\left(\frac{\pi+2}{2\pi}\right)$ **D)** $\frac{\mu_o i}{4\pi a}\left(\frac{2\pi}{\pi+2}\right)$

29. Consider an infinite combination of resistors as shown in the figure. What is the effective resistance between points A and B for the shown combination:



A) $\left(\frac{\sqrt{3}+1}{2}\right)R$ **B)** $\left(\frac{\sqrt{7}+1}{2}\right)R$ **C)** $\frac{3}{2}R$ **D)** $\left(\frac{\sqrt{5}+1}{2}\right)R$

30. A small circular loop of area A and resistance R is fixed on a horizontal xy-plane with the center of the loop always on the axis $\hat{n}$ of a long solenoid. The solenoid has m turns per unit length and carries current I counter clockwise as shown in the figure. The magnetic field due to the solenoid is in $\hat{n}$ direction. List-I gives time dependences of $\hat{n}$ in terms of a constant angular frequency ω List-II gives the torques experienced by the circular loop at time $t = \frac{\pi}{6\omega}$. $Let\ \alpha = \frac{A^2\mu_0^2 m^2 I^2 \omega}{2R}$



List-I		List-II	
I	$\left(\sin\omega t\hat{j}+\cos\omega t\,\hat{k}\right)$	**P**	**0**
II	$\left(\sin\omega t\hat{i}+\cos\omega t\,\hat{j}\right)$	**Q**	$-\frac{\alpha}{4}\hat{i}$
III	$\left(\sin\omega t\hat{i}+\cos\omega t\,\hat{k}\right)$	**R**	$\frac{3\alpha}{4}\hat{i}$
IV	$\left(\cos\omega t\hat{j}+\sin\omega t\,\hat{k}\right)$	**S**	$\frac{\alpha}{4}\hat{j}$
		T	$-\frac{3\alpha}{4}\hat{i}$

Which one of the following options is correct?

A) $I\to Q,\ II\to P, III\to S, IV\to T$ **B)** $I\to S,\ II\to T, III\to Q, IV\to P$

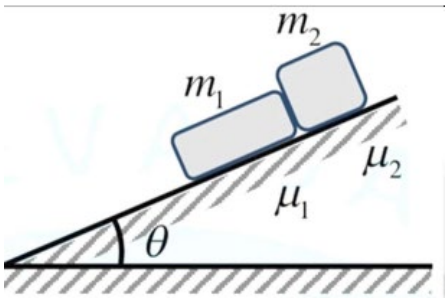
C) $I\to Q,\ II\to P, III\to S, IV\to R$ **D)** $I\to T,\ II\to Q, III\to P, IV\to R$

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

31. Two blocks A and B of masses m1 and m2 are placed in contact with each other on an inclined plane as shown. The co – efficient of friction of block A and B with inclined surface is $\mu_1 \text{ and } \mu_2$ respectively with $\mu_1 > \mu_2$ .The angle of inclination is θ .Mark the correct option:



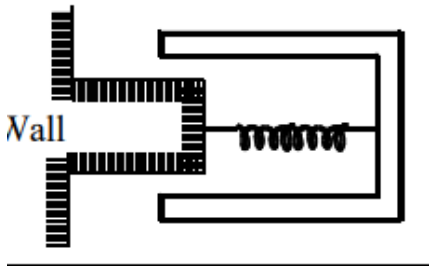
A) If both the blocks do not slip on the incline, then $\tan\theta \le \frac{\mu_1 m_1 + \mu_2 m_2}{(m_1 + m_2)}$

B) If both blocks slip together on incline, then their acceleration is $g\sin\theta - \left(\frac{\mu_1 m_1 + \mu_2 m_2}{m_1 + m_2}\right) g\cos\theta$

C) If both blocks slip on incline, then the normal force between the blocks is always zero.

D) Under a situation that $\mu_1 < \mu_2$ and both blocks slipping on incline, the normal force between the blocks will become zero.

32. Consider the situation shown in diagram in which a hollow cylindrical box of mass m, open at one end is connected to an ideal spring. Initially spring is compressed and box having mass m is kept stationary. Spring is attached with box but not attached with wall and initially held at rest. Now box is released. Choose INCORRECT options



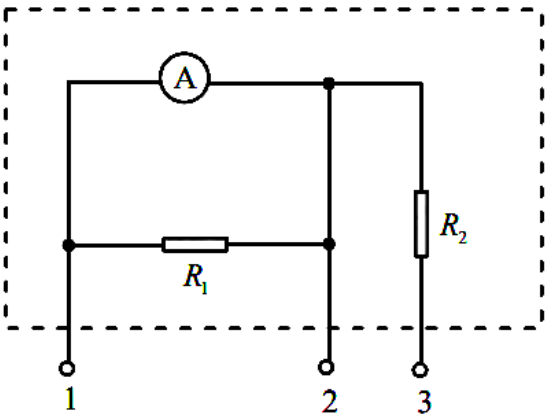
A) Work done by wall on box plus spring system is zero

B) Work done by wall on spring plus box system is positive

C) Change in kinetic energy of box plus spring system is zero

D) Work done by wall on box plus spring system is zero but change in kinetic energy of this systems is positive, so work energy theorem fails in this case

33. As shown in the figure, the range of ammeter A is 0.6A, the dial is evenly divided into 30 small grids, each small grid represents 0.02A; the resistance R_1 is equal to the resistance of the ammeter, which is equal to 100Ω The resistance of R_2 is equal to 2 times the resistance of the ammeter.



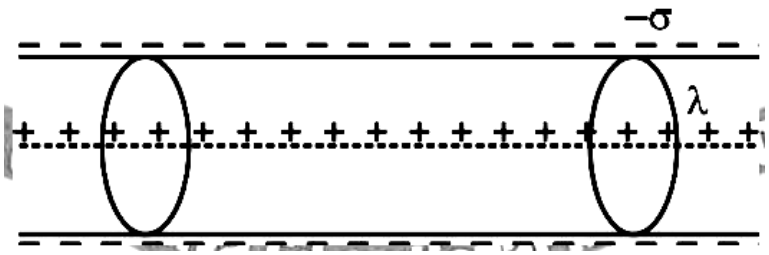
A) When connecting terminals 1 and 2 to the circuit, as an ammeter, each small grid represents 0.04A.

B) When connecting terminals 1 and 2 to the circuit, as an ammeter, each small grid represents 0.02A.

C) When connecting terminals 1 and 3 to the circuit, as an voltmeter, each small grid represents 10V.

D) When connecting terminals 1 and 3 to the circuit, as an voltmeter, each small grid represents 5V.

34. Figure shows a long thread along the axis of a long hollow cylinder. The uniform charge per unit length of thread is λ while uniform surface charge density of the cylinder is $-\sigma$. The radius of the cylinder is R. Then correct option among the following is/are



A) Total flux associated with a coaxial cylinder of finite length and radius twice that of the charged cylinder is zero, if the magnitude of $\sigma = \frac{\lambda}{2\pi R}$

B) Total flux associated with a coaxial cylinder of finite length and radius twice that of the charged cylinder is zero, if the magnitude of $\sigma = \frac{\lambda}{4\pi R}$

C) Total flux associated with a coaxial cylinder of finite length and radius half that of the charged cylinder is independent of σ

D) Total flux associated with a coaxial cylinder of finite length and radius half that of the charged cylinder is zero and independent of σ

SECTION – III

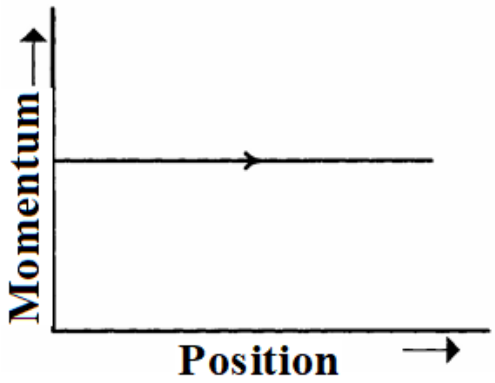
(PARAGRAPH QUESTIONS TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions and based on the other paragraph 2 multiple choice questions have to be answered. Each of these questions has four choice (A) , (B) , (C) and (D) out of which ON:Y ONE is correct

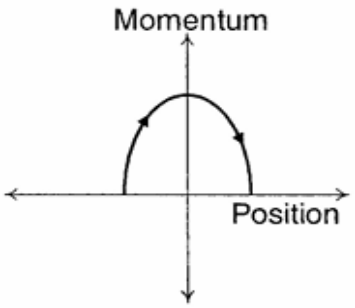
Passage-1

Phase space diagrams are useful tools in analyzing all kinds of dynamical problems. They are especially useful in studying the changes in motion as initial position and momentum

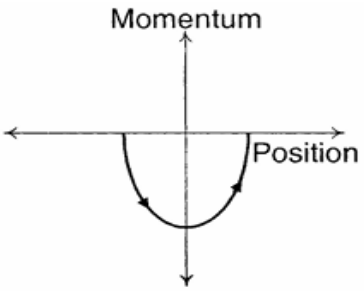
are changed. Here we consider some simple dynamical systems in one-dimension. For such systems, phase space is a plane in which position is plotted along horizontal axis and momentum is plotted along vertical axis. The phase space diagram is x(t) vs. p(t) curve in this plane. The arrow on the curve indicates the time flow. For example, the phase space diagram for a particle moving with constant velocity is a straight line as shown in the figure. We use the sign convention in which position or momentum upwards (or to right) is positive and downwards (or to left) is negative.



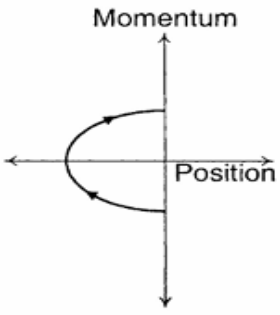
35. The phase space diagram for a ball thrown vertically up from ground is



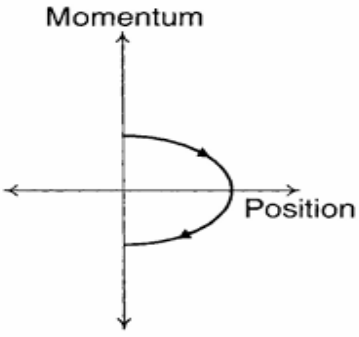
A)



B)

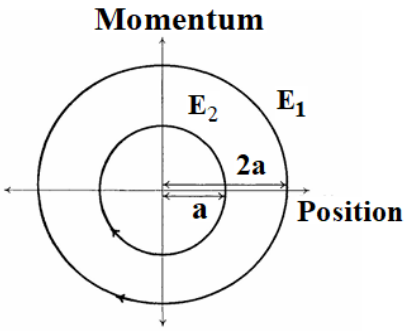


C)



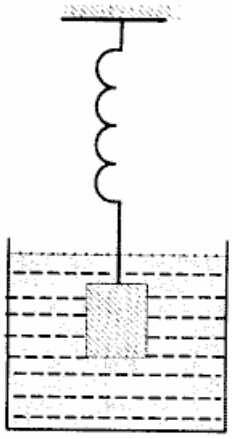
D)

36. The phase space diagram for simple harmonic motion is a circle centered at the origin. In the figure, the two circles represent the same oscillator but for different initial conditions, E_1 *and* E_2 are the total mechanical energies respectively. Then

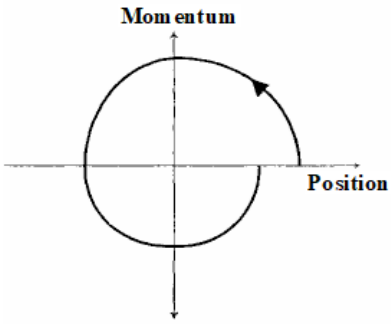


A) $E_1 = \sqrt{2}\,E_2$ **B)** $E_1 = 2\,E_2$ **C)** $E_1 = 4\,E_2$ **D)** $E_1 = 16\,E_2$

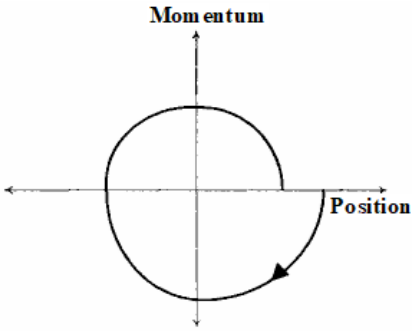
37. Consider the spring-mass system, with the mass submerged in water, as shown in the figure. The phase space diagram for one cycle of this system is



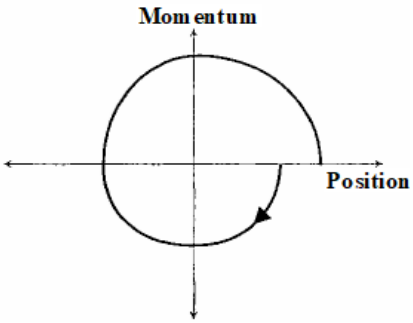
A)



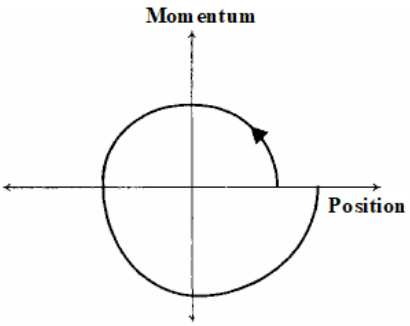
B)



C)



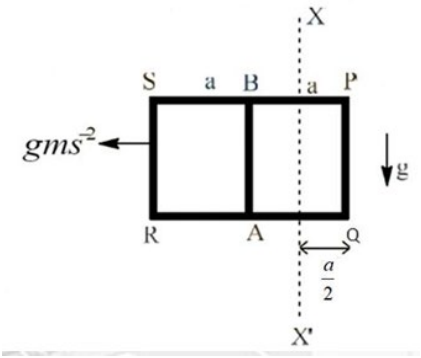
D)



Passage-2

A conducting loop is shown figure, it is made of seven identical wires each of length a, Resistivity 'ρ', area of cross-section S_0. The frame lies in a vertical plane. XX ' is an imaginary boundary separating space in two parts. Right to XX', a uniform gravitational field g exists (figure) whereas no gravitational field is present left of XX '. Instantaneously frame is accelerating in leftwards with acceleration gm/s^2, e and m are

electronic charge and mass of electron respectively. Ignore inductive and capacitive properties of loop. Instantaneously



38. The magnitude of potential difference between points P and Q is

A) $\frac{7}{15}\frac{mga}{e}$ **B)** $\frac{mga}{e}$ **C)** $\frac{11}{15}\frac{mga}{e}$ **D)** $\frac{15}{7}\frac{mga}{e}$

39. Current in branch RS is

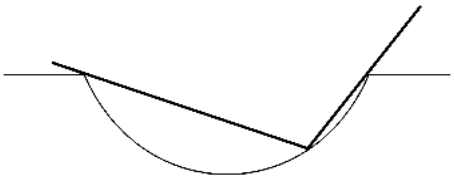
A) $\frac{4mgS_0}{15\rho e}$ **B)** $\frac{mgS_0}{15\rho e}$ **C)** $\frac{4mgS_0}{\rho e}$ **D)** $\frac{15mgS_0}{4\rho e}$

SECTION –IV

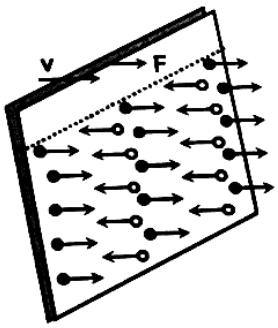
(INTEGER ANSWER TYPE)

This section contains **7 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

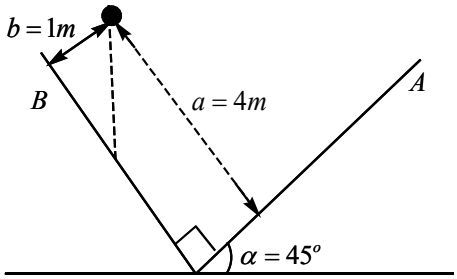
40. A semi-cylindrical groove of radius r=20 cm is made on a horizontal floor. An ant wants to cross the groove. A boy decides to help the ant making a bridge consisting of straight wire segments. But all the wires available are of length $l = 32cm$, so the boy rigidly connects two wires at right angle and places the bridge in the groove as shown in the figure. If the ant can crawl up a wire segment at speed $\upsilon = 0.5cm/s$ and down a wire segment at speed 2υ, in what minimum time(T) in seconds can the ant cross the groove with the help of this bridge? The left part of the bridge is just inside the groove. Find the value of x. If the time (T)is $10x$.



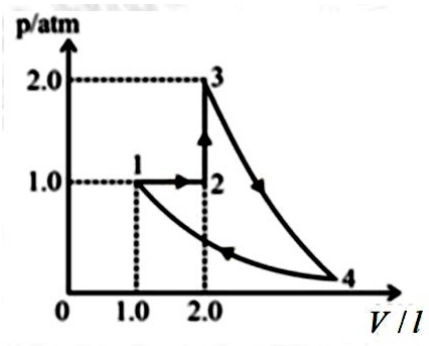
41. A thin heavy metal plate is being bombarded by a perpendicular beam of gas particles from both sides as shown in the figure. The solid dots are representing the molecules hitting from left side and the faint dots are the molecules hitting from right side. The mass of these gas particles is $m = 10^{-26}kg$ and velocity before hitting $v_0 = 5m/s$. Volume density of the gas particles on both sides is $n = 10^{25}\,per\,m^3$ .Each beam has an area $A = 1m^2$ and the collisions are perfectly elastic. What is the external force F (in newton) required to move the plate with a constant velocity v = 2 m/s.



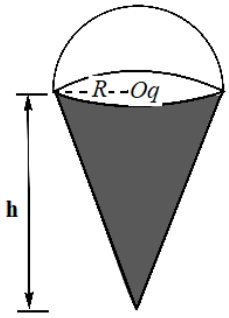
42. Two boards have been fixed at right angles to one another. Their line of touching is horizontal. An elastic ball is released at a point at distance a=4m from the plane A and b=1m from B. On the average, how many times does the ball bounce against wall B for each time it bounces against wall A? collisions are absolutely elastic



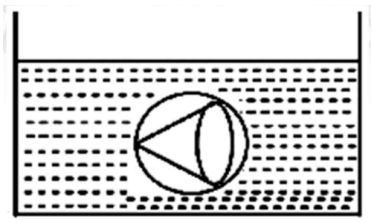
43. 3.2g of oxygen starts from the initial state $1\left(p_1 = 1.0\,atm, V_1 = 1.0L\right)$, reaches state 2 $\left(V_2 = 2.0L\right)$ through an isobaric process, and then passes through the isochoric process. The process reaches state $3\left(p_3 = 2.0atm\right)$, and then reaches state 4 through an adiabatic process. The temperature of state 4 is exactly the same as the temperature of state 1. Finally. It return to state 1 through an isothermal process, forming a cyclic process. $\left(Take\, P_4 = 0.0157\,atm, V_4 = 64L\right)$ The efficiency of cyclic process is $1-\frac{3x\ln 2}{17}$ then x value is



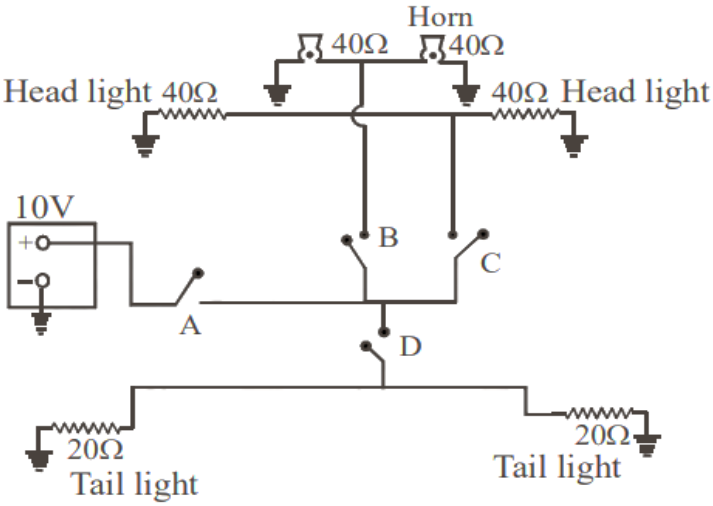
44. A charge q is surrounded by a closed surface consisting of an inverted cone of height h and base radius R, and a hemisphere of radius R as shown in the figure. The electric flux through the conical surface is $\frac{nq}{6\varepsilon_0}$ (in SI units). The value of n is _____.



45. A solid cone is tightly fitted inside a thin massless spherical shell of radius R = 6m. The semivertex angle of the cone is 30° and density $\frac{32}{9} g/cm^3$ .The system is kept immersed inside water so that the axis of the cone lies in a horizontal plane is shown. If it is released from this position, net torque about a space point at vertex of the cone immediately after the release is x terms of $36\pi \times 10^4 Nm$. Find the value of x $\left[g = 10 m/s^2, \rho_{water} = 1 g/cm^3 \right]$



46. Figure shows an automobile circuit. If the power (in watt) dissipated by the automobile circuit when switches A, B, C and D are all closed is 4n. Then find the value of n



MATHEMATICS: Max.Marks: 80S

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

---

47. **If** $(x-a)\cos\theta + y\sin\theta = (x-a)\cos\phi + y\sin\phi = a, \tan\frac{\theta}{2} - \tan\frac{\phi}{2} = 2e$ and θ, ϕ are unequal angles less than $360°$,then y^2 is equal to:

A) $2ax - (1+e^2)x^2$ **B)** $2ax - (1-e^2)x^2$ **C)** $2ax + (1-e^2)x^2$ **D)** $2ax + (1+e^2)x^2$

48. Consider a function f which satisfies $f(x,y)=f(2x+2y,2y-2x)$.Now consider the function $g(x)=f(2^x,0)$ The period of $g(x)$ is

A) 4 **B)** 8 **C)** 12 **D)** 16

49. Tangents are draws to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ and the circle $x^2+y^2=a^2$ at the points where a common ordinate cuts them (on the same side of the x – axis). Then the greatest acute angle between these tangents is given by:

A) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$ **B)** $\tan^{-1}\left(\frac{a+b}{2\sqrt{ab}}\right)$ **C)** $\tan^{-1}\left(\frac{2ab}{\sqrt{a-b}}\right)$ **D)** $\tan^{-1}\left(\frac{2ab}{\sqrt{a+b}}\right)$

50. A tetrahedron has vertices at O(0,0,0), A(1,2,1), B(2,1,3) and C(-1,1,2). Then, the angle between the faces OAB and ABC will be

A) 90^0 **B)** $\cos^{-1}\left(\frac{19}{35}\right)$ **C)** $\cos^{-1}\left(\frac{17}{31}\right)$ **D)** 30^0

51. A function $f(x)$ satisfies the condition, $f(x)=f'(x)+f''(x)+f'''(x)+.....\infty$ where $f(x)$ is an indefinitely differentiable function and dash denotes the order of derivatives. If $f(0)=1$, then $f(x)$ is

A) $e^{\frac{x}{2}}$ **B)** e^x **C)** e^{2x} **D)** e^{4x}

52. If $G(x,t)=\begin{cases}x(t-1), & when\, x\le t\\ t(x-1), & when\, t<x\, and\, if\, t\, is\, continuous\, funciton\, of\, x\, in[0,1]\end{cases}$

Let $g(x)=\int_0^1 f(t)G(x,t)dt$. Then which is incorrect

A) $g(0)+g(1)=0$ **B)** $g(0)=0$

C) $g(1)=1$ **D)** $g''(x)=f(x)$

53. The equation $(1+a)x^2+2a^2x+a^2+b^2-1=0$ has roots of opposite sign, if $a+ib\ lies,(a>-1)$:

A) On straight line $x+y=1$

B) Inside a circle of centre (0,0) and radius '1'

C) On a parabola of vertex (0,0) and focal length '1'

D) None of the above

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

54. The sequence $\{a_n\},n\in N$ satisfies $a_1=1$ *and* $5^{a_{n+1}-a_n}=1+\frac{1}{n+\frac{2}{3}}$. Then

A) $[a_{501}]=3$ **B)** $[a_{207}]=3$ **C)** $[a_{223}]=4$ **D)** $[a_{625}]=4$

55. Identify the statement(s) which is/are true?

A) The order of differential equation $\sqrt{1+\frac{d^2y}{dx^2}} = x$ *is* 1

B) Solution of the differential equation $xdy - ydx = \sqrt{x^2+y^2}dx$ *is* $y+\sqrt{x^2+y^2} = Cx^2$

C) $\frac{d^2y}{dx^2} = 2\left(\frac{dy}{dx} - y\right)$ is differential equation of family of curves $y = e^x\left(A\cos x + B\sin x\right)$

D) The solution of differential question

$\left(1+y^2\right)+\left(x-2e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0$ *is* $xe^{\tan^{-1}y} = e^{3\tan^{-1}y} + k$

56. If a, b, c, d are positive rational numbers such that $a \pm b$ *and* $c \pm i\sqrt{d}$ satisfy the equation

$\left(\frac{x}{2}-3\right)^4 = 1-\left(\frac{x}{2}-4\right)^4$, then

A) $a-b+c-d$ = number of zeroes at the end of 27!

B) None of the number a, b, c, d is composite

C) $a+b+c+d$ is even

D) $a+b+c+d$ is odd

57. Let $n \in N, n \geq 4$ and $P = \prod_{r=0}^{n} {}^nC_r$, then

A) $P > \left(\frac{2^n}{n+1}\right)^{n+1}$ **B)** $P < \left(\frac{2^n}{n+1}\right)^{n+1}$ **C)** $P < \left(\frac{2^n-2}{n-1}\right)^{n-1}$ **D)** $P < \left(\frac{2^n-2}{n-1}\right)^{n}$

SECTION – III

(PARAGRAPH QUESTIONS TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions and based on the other paragraph 2 multiple choice questions have to be answered. Each of these questions has four choice (A) , (B) , (C) and (D) out of which ON:Y ONE is correct

Passage – 1

Let $a = \cos^{-1}\cos 20, b = \cos^{-1}\cos 30$ *and* $c = \sin^{-1}\sin\left(a+b\right)$, then

58. The largest integer x for which $\sin^{-1}\left(\sin x\right) \geq \left|x-\left(a+b+c\right)\right|$ is

A) 1 **B)** 2 **C)** 3 **D)** 4

59. If $5\sec^{-1}x + 10\sin^{-1}y = 10\left(a+b+c\right)$ then the value of $\tan^{-1}x + \cos^{-1}\left(y-1\right)$ is

A) $\frac{\pi}{2}$ **B)** $\frac{\pi}{4}$ **C)** π **D)** 0

Passage – 2

If a circle C_0 with radius 1 unit touches both the axes and as well as line $\left(L_1\right)$ through $P\left(0,4\right)$, L_1 cut the x-axis at $\left(x_1,0\right)$. Again a circle C_1 is drawn touching x-axis, line L_1

and another line L_2 through point P. L_2 intersects x-axis at $(x_2,0)$ and this process is repeated n times

60. The value of x_2 is

A) 3 B) 4 C) 15/2 D) 17/2

61. The centre of circle C_2 is

A) (3,1) B) $\left(\frac{15}{2},1\right)$ C) $\left(\frac{31}{4},1\right)$ D) $\left(\frac{17}{2},1\right)$

62. The $\lim_{n\to\infty}\frac{x_n}{2^n}$ is

A) 4 B) 0 C) 1 D) 2

SECTION –IV

(INTEGER ANSWER TYPE)

This section contains **7 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

63. A function $f:(-1,1)\to R$ is given by $f(x)=\frac{1}{1+x}+\frac{2x}{1+x^2}+\frac{4x^3}{1+x^4}+\frac{8x^7}{1+x^8}+.....\infty$

Let i be the number of solutions of the system

$y=f(x)$

$(x-1)^2+y^2=2$

Let α_r be the roots of the system. Then the value of $\sum_{r=1}^{i}[\alpha_r]$ is

64. Deepak has $2n$ number of fruits out of which *n* of them are identical and remaining n are distinct. If the total number of ways he can distribute these fruits to his students Bhavesh and Dhinesh such that both of them will receive equal number of fruits is 16 then n is equal to_____.

65. A person can take one step or two steps or three steps in forward direction at a time. If "P" is the number of distinct ways he can reach at the end of 8 steps then the value of $\sqrt{P}$ is

66. Let A_n *and* B_n be square matrices of order 3, which are defined as

$A_n=\left[a_{ij}\right]$ *and* $B_n=\left[b_{ij}\right]$ where $a_{ij}=\frac{2i+j}{3^{2n}}$ *and* $b_{ij}=\frac{3i-j}{2^{2n}}\forall i$ *and* $j, 1\le i,j\le 3,$ if

$l=\lim_{n\to\infty}T_r\left(3A_1+3^2A_2+3^3A_3.....+3^nA_n\right)$ and $m=\lim_{n\to\infty}T_r\left(2B_1+2^2B_2+2^3B_3.....+2^nB_n\right)$, then the value of $\frac{l+m}{3}$, where $T_r(A)$ represent trace of matrix A

67. The chord AC of the parabola $y^2 = 4ax$ subtends an angle of 90^0 at points B &D on the parabola. If points A, B, C and D are represented by $\left(at_i^2, 2at_i\right)$ where i=1,2,3,4 respectively, then the value of $\left|\frac{t_2 + t_4}{t_1 + t_3}\right|$ is

68. The number of real solutions to the equation $3x - 7 = \left[x^2 - 3x + 2\right]$ is _____. $[x]$ denotes greatest integer $\leq x$

69. If $0 \leq \theta \leq \pi$ and the system of equations

$x = (\sin\theta)y + (\cos\theta)z,$

$y = z + (\cos\theta)x$

$z = (\sin\theta)x + y$

has a non-trivial solution, then $\frac{8\theta}{\pi}$ is equal to